QFF 2026 — Processor A/B Benchmark Configuration

Challenge Kit — Configuration for Final Approval

Purpose: This document consolidates the processor and noise-model information currently present in the QFF challenge materials and identifies the items that must be frozen in the official Challenge Kit so every participant uses the same benchmark.

IMPORTANT STATUS: Processor A and Processor B maps in this document have now been corrected to match the simulation configurations supplied for the hackathon. Processor A = 5 qubits; Processor B = 9 qubits. The exact noise parameters are NOT verified by the processor-map screenshots and must be frozen separately in the official Challenge Kit.

1. Issues identified in the current materials

- Processor A is described in the executive summary as a 5-qubit linear-chain baseline, while the supplied Processor A diagram shows a 5-node branching topology.
- Earlier written materials described Processor B differently, while the supplied simulation configuration explicitly contains 9 nodes.
- The Starter Guide contains an illustrative noise model and explicitly states that its values must be replaced by the official challenge assumptions.
- Different challenge materials contain different example CX-noise values (approximately 0.5% in the executive summary and p=0.005 in one detailed problem statement, while the Starter Guide's illustrative code uses p=0.01).
- Therefore, participants should not infer processor maps or official noise parameters from the examples. The Challenge Kit must be the single source of truth.

2. Source-supported baseline

Simulation-derived authoritative configuration: Processor A uses 5 physical qubits. Processor B uses 9 physical qubits. These values and their coupling maps are taken directly from the simulation configurations supplied for this hackathon. The earlier written description of Processor B is inconsistent with the simulation and should not be used for the benchmark.

3. Processor A — exact simulation configuration

The following is the exact 5-qubit Processor A configuration from the simulation output supplied for the hackathon:

Qubit Diagram position Connections q0 Node 0 q1

q1 Node 1 (central) q0, q2, q3 q2 Node 2 q1

q3 Node 3 q1, q4

q4 Node 4 q3

Exact simulation coupling edges (unique/undirected): (0,1), (1,2), (1,3), (3,4).

Verification: The supplied simulation explicitly defines PHYSICAL_QUBITS = 5 and the directed edge list [(0,1),(1,0),(1,3),(3,1),(1,2),(2,1),(4,3),(3,4)]. Its unique physical connections are therefore (0,1), (1,2), (1,3), and (3,4).

4. Processor B — exact simulation configuration

The following is the exact 9-qubit Processor B configuration from the simulation output supplied for the hackathon. This resolves the earlier earlier written processor-count inconsistency: the simulation uses 9 physical qubits.

Qubit Diagram position Connections q0 Node 0 q1, q8

q1 Node 1 q0, q2, q6 q2 Node 2 q1, q3, q5, q6 q3 Node 3 q2, q4

q4 Node 4 q3, q5

q5 Node 5 q2, q4, q6 q6 Node 6 q1, q2, q5, q7, q8 q7 Node 7 q6

q8 Node 8 q0, q6

Exact simulation coupling edges (unique/undirected): (0,1), (1,2), (1,6), (2,3), (2,6), (2,5), (3,4), (4,5), (5,6), (6,7), (6,8), (8,0).

Verification: The supplied simulation explicitly defines PHYSICAL_QUBITS = 9 and the directed edge list contains both directions for 12 physical connections. The exact unique connections are listed above. The simulation therefore uses all q0–q8; no two nodes should be removed.